

Sai Kiran Kalyankar

Enthusiast Problem Solver, MS CS Grad with 5+ years of Experience

+1 3132638812

kalyankarsai@gmail.com

sai-kiran-kalyankar

kalyankarsai

Windows Dev

Android Dev

Full Stack Dev

4500 Cass Ave, UT Apt 806, Detroit, MI 48201, USA

Summary

Highly skilled and accomplished Senior Software Engineer with over 5 years of expertise in full-stack development. Proficient in various technologies including Java/J2EE, Spring MVC, Hibernate, and JavaScript (jQuery, Angular, React). Experienced in Microservice architecture using Spring MVC, Spring Boot, NestJS, and ASP.NET, with configuration using Docker and Jenkins for continuous integration and build processes. Skilled in BDD and TDD using JUnit and Mockito, and experienced in cloud environments such as GCP, PCF, AWS, and Azure. Extensive experience working on agile teams to deliver high-impact, scalable solutions in cloud platforms, while advocating for best practices in software engineering, usability, and maintainability. Proficient in front-end technologies like React and Angular, and experienced in DevOps practices and AI/ML integration.

Key Highlights

- > 5+ years of Java/J2EE development expertise, including Java Collections Framework, RESTful APIs, JPA, and JUnit.
- > 4+ years of Spring Boot experience in building microservices and scalable backend solutions.
- > 3+ years of cloud experience in Pivotal Cloud Foundry (PCF), contributing to cloud-native applications.
- > 1+ years of cloud experience in Google Cloud Platform (GCP).
- > 2+ years of cloud experience in AWS and Azure.
- > Strong understanding of DevOps tools like Git, Docker, Jenkins, and Gradle, with 4+ years of experience supporting CI/CD pipelines.
- > 2+ years of relevant experience using Hadoop Ecosystem tools and their architecture.
- > 1+ years of AI/ML experience, including Google Vertex AI hosted on Huggingface models, integrating AI/ML models into cloud applications.
- > 1+ years of experience in .Net Programming.
- > Expertise in creating responsive front-end applications from wireframes using Angular and React.
- > Experienced working on full stack web, mobile, server, and database using multiple programming languages.
- > Proven ability to work in agile environments with two-week iterations, ensuring timely delivery and adherence to best practices.
- > Experience in the full SDLC including Scrum/Agile/Kanban and mob/pair programming (Xtreme Programming) methodologies, including experience applying Test-Driven Development(TDD), BDD, and Static Code analysis to improve quality and reliability.
- > Advocate of DevSecOps, collaborating on the support and enhancement of DevOps pipelines for continuous delivery.

Competencies

Languages : Java, TypeScript, Python, C#, C/C++, PHP, JavaScript, Ajax, jQuery, Kotlin, Groovy, GoLang, HTML, CSS, Sass, SQL, NoSQL, GraphQL, TeX

Frameworks : Angular, Spring-Boot, Hibernate, NestJS, TypeORM, .Net 7, NServiceBus, Jest, JUnit, Mockito, xUnit, Node.js, FastAPI, Flask, Hadoop, Hive

DevOps and API Tools : Git, SVN, Docker, Kubernetes, AWS, Azure, PCF, GCP, Bash, Shell, GitLab, Bitbucket, Jenkins, Sonar, DAST, Swagger, Postman, Splunk

Tools : MySQL, MSSQL, OracleDB, Postgres, MongoDB, SQLite, Impala, oozie, Figma, VS Code, Visual Studio, Android Studio, PyCharm, IntelliJ, NetBeans, Eclipse

Libraries : React, Redux, handlebarsjs, pug, pandas, NumPy, Matplotlib, scikit-learn, CUDA, YOLO, NLP, NLTK, OpenCV, OpenGL, WebGL, ThreeJS, OpenAI

Other Skills : 3D Computer Vision and Graphics, Machine Learning, Mobile, Android, Windows App development, UNIX, Powershell, Salesforce

Work Experience

Research Developer Lead

Lipidomics Department-Wayne State University, Detroit | Sept 2023 –Present

- > Working on building a Full Stack Web Application Solution for the Lipidomics Department at WSU.
- > Leading the application development from scratch and continuously delivering features for both backend and frontend.
- > Utilized the internal platform CITADEL (C&IT Application DELivery) for deployment.

ReactJS Python FastAPI SQLAlchemy MySQL GitLab JavaScript Bootstrap nginx bash Docker Inventory Database Cost Calculator Billing

Software Engineer Intern

Amrock-Rocket Companies, Detroit | May 2023–Aug 2023

- > Full Stack Developer for Technocrats Platform Team to migrate monolith internal product(Atlas) to Cloud microservices using Azure and AWS
- > Designed and implemented Angular frontend and C#/ASP.NET backend, using CQRS event-driven architecture, Pub/Sub messaging model
- > Developed APIs along with regular code reviews, ensuring high quality and adherence to 100% compliance with security coding standards
- > Assisted the team in bug fixing, addressing feature requests, resolving security vulnerabilities, writing unit tests, integration tests for the code.
- > Utilized GCP Cloud Services

- > **Achievement :** Received **Amaze Recognition** for creating team's GraphQL Service -**Rocket Companies Hackweek Q2 - June 2023**

Angular 15 C# .Net 7 ASP.Net MS SQL Server Docker AWS Azure DevOps CQRS GCP PCF GraphQL NServiceBus Monorepo Snyk Splunk

Continuing Education in USA

Master's Degree-Wayne State University, Detroit, MI | August 2022 –May 2024

Analyst, Specialist, Application Development

DBS Bank - Development Bank of Singapore-Hyderabad | Feb 2022–Aug 22

- > Java Full Stack developer, for Consumer Banking for Investment and Insurance Project - FinTech using Frontend 20% and Backend 80%.
- > Senior Backend Developer at SAILOR team (C2E) worked on Gamification Project(NavGames - To teach about finances in a fun way, by introducing a series of mini-games) using RESTful Microservices with Spring Boot.
- > Ownership of Product development for Internet Banking and Mobile Banking about financial planning and wealth management for Digibank.
- > Collaborated with cross-functional teams, testing using GoMock, documented tech and design specifications.
- > Presented project proposals and progress updates to higher management, conducted code reviews, and mentored junior developers, with Successful Go-Live, increasing user engagement by 30%.
- > Microservices deployed and scaled on PCF Cloud Services.

- > **Achievement :** **DBS TECH India Annual Award** Top Engineering Team - SAILOR - C2E. -**Aug 2022**

Spring-boot Micro-service Java 11 Groovy GoLang MariaDB REST GraphQL Pivotal Cloud Foundry Product Dev Gamification Figma Unit Test Redis Maven Gradle SonarQube Bitbucket Jira postman Docker Kubernetes Jenkins NAV-Planner GCP HLD LLD Mentorship

Technology Analyst

Infosys - Client : Verizon-Hyderabad | April 2021–Feb 2022

JavaScript Full Stack developer as part of VZ Network Transparency ART(Service Assurance), Designed, developed, and maintained RESTful API endpoints.

Led developing microservices including design and implementation of new products and features while enhancing existing product suite.

Using an asynchronous distributed messaging and streaming platform(Apache Pulsar similar to Kafka) with security standards having Gov. and GIS data.

Using SAFe agile framework, collaborating with business and research teams in story grooming, reviewing story/acceptance criteria and Iteration planning

Utilised AWS EKS, RDS(Postgres), Frontend 20%, Backend 80%, Achieved a 95% incident resolution rate through dedicated service ownership

Deployed and Maintained microservices on Google Cloud Platform with High availability to support the maintenance of services developed.

ReactJS

NestJS

Microservice

Springboot

NodeJS

TypeScript

TypeORM

JavaScript

SCSS

Pug

handlebarsjs

Kubernetes

Docker

Jenkins

Postgres

MongoDB

Jest

PCF

GCP

CI/CD

pulsar-client

Sonar

Fortify

TwistLock

Echo

Kirke

notifications

Engineer - Technology

Virtusa - Client : Openreach - BT(British Telecom), Lloyds Banking Group-Hyderabad | Dec 2017–April 2021

Full Stack Java Developer, LAMP Stack Backend Developer using DevOps principles to add new functionality to LLUMS - Openreach’s products.

Developed application components in an Agile environment utilizing TDD to track E2E fiber build activity for SiteTracker (integrated with Salesforce) Squad, using Hadoop as a Service (HaaS) and configuring Workflow Orchestration.

Participated in major project migration, developing data models, pages, migration scripts, and optimizing systems to boost app performance by 50%.

Performed database, SQL tuning and optimization for MySQL and Postgres achieving 99% uptime and reducing response time by 15%.

Deployed the spring microservice applications on PCF and scaled as needed using PCF services.

Automated uploaders, inventory flows using SharePoint Excel File IO and SMTP notifications, enhancing monitoring and reducing manual effort by 30%

Developed scripts to automate daily, weekly, and monthly cronjobs, including report generation and real-time batch-feed processing using ETL and logs.

Created Inventory Self Service Reports and billing reports using Spring JDBC and IBM MQ Messaging Queue, including configuration and administration.

Involved in continuous enhancements using Frontend Angular 7, including functional & non-functional requirements and fixing of production issues.

Delivered results in complex cross-functional teams, presented, collaborated, and added value through documentation, debugging, test support and RCA.

Participated in peer code reviews, mentored other engineers, providing comments and suggestions.

Achievements : Appreciation and tribe recognition for the work in “Agile SiteTracker squad” at BT -Nov 2020

Certificate of Appreciation for exemplary performance and contribution to “LLUMS team” at Virtusa.-Mar 2019

Client Appreciation with virtual client love badge. -Dec 2018

J2EE

Spring

Hibernate

JUnit

Mockito

MVC

Spring-Boot

Bootstrap

JS

Pivotal Cloud Foundry

jQuery

PHP

MySQL

Unix

Shell

bash

Git

SVN

REST

Self-Service Views

cron

GitLab

Docker

Citrix

IBM MQ

SOAP

Hadoop

sqoop

hive

Cloudera HUE

oozie

Informatica

OLTP

OLAP

ETL

Other Projects

Detroit Auction Shippers

Independent Project-Detroit | Nov 2023

Created Full Stack Web App for an Auction Shipper Company in Detroit for Real-time use.

Integrating GenAI using Spring AI, OpenAI, LLM

ReactJS

Spring Boot

MySQL

Git

SPA

Bootstrap

SMTP

HTML

CSS

AWS

EC2

S3

RDS

Business and Architect Solution

Protfolio QA AI App

Independent Project- | June 2024

WebAPP AI Chatbot : AI App for answering questions about My portfolio.

Gemini AI

RAG

Vertex AI

Web

Chatbot

Huggingface

Pictionary Detector AI App

Independent Project- | June 2024

Developed an AI-based Flutter app to detect and predict objects drawn during a Pictionary game in real time

Gemini AI

RAG

Vertex AI

Android

Web

Flutter

Text Spam Detection

Course Project-Wayne State University | Dec 2023

Using machine learning techniques processed data through essential libraries, conducted exploratory analysis, and performed feature engineering, Constructed a prediction model achieving accurate classification of spam texts.

ML

Support Vector Machines (SVM)

Multinomial Naive Bayes

Gaussian Naive Bayes

Logistic Regression

Random Forest

AdaBoost

3D Graphics Renderer using OpenGL, C++

Course Project-Wayne State University | Apr 2023

Developed a 3D graphics renderer leveraging OpenGL and C++ to create immersive visualizations. Implemented scene composition, lighting effects, and texture mapping for realistic rendering. Applied geometry transformations, rendering techniques, and mathematical algorithms to achieve high-quality graphics.

Geometry transformations

Rendering pipeline

Texture mapping

Lighting techniques

mesh

Ray tracing

OpenGL programming

Sentiment Analysis on Political Tweets

Course Project-Wayne State University | Apr 2023

The aim of this project is to perform sentiment analysis on political tweets during the 2020 United States presidential election.to analyze the sentiments of tweets related to the election, identify the patterns, and explore different techniques of data mining from simple classification models to more advanced models.

Data Mining

NLP

NLTK

scikit-learn

Sentiment Analysis

Vader

Random Forest

Logistic Regression

RNN

LSTM

Choropleth

The Big Five Personality Analysis

Course Project-Wayne State University | Dec 2022

> The Project goal is to use unsupervised clustering techniques to identify natural groupings or patterns within the data. These groupings may reveal relationships between the different personality traits and potentially provide insights into the underlying structure of personality using the Big Five Personality traits dataset and custom inputs

PythonGoogle colabIPIP datasetsklearnK-MeansPCAData PreprocessingData visualizationIntelligent Systems

Instant NGP

Course Project-Wayne State University | Dec 2022

> Implementation project on Paper about Instant Neural Graphics Primitives with a Multiresolution Hash Encoding by NVIDIA, Using own custom real-time data

Machine learning3D computer visionneural-networkcomputer-graphicsrealtimepytorchgpucudapythoncolmapGANVAEsigned-distance-functionsnerf3d-reconstructionfunction-approximationreal-time-renderingHPCDeeplearning

Automatic Curation of Images

Internship-Teletext Holidays | Feb–Mar 2017

> Automatic curation of images based on the user’s requirement primarily in order to check if descriptions can match the image’s contents and to validate images against their respective descriptions primarily to curate the hotel images of the website as per the user’s preferences.

> Real-time scene classification using Caffe’s deep learning framework and Object Detection using YOLO’s CNN over Machine Learning Techniques.

Machine LearningCUDAcuDNNDFlaskCaffeYOLOOPENCVBootstrapPython

Authentication Using Graphical Passwords

Major Project-Vardhaman College | Jan–Apr 2017

> In this project an authentication portal has been designed where a Multi-factor authentication system is built on top of Puzzle technology to make passwords easier for people to use and therefore more secure.

> With an Add-On feature of Face Recognition (using Haar Cascade Classifier) based on Machine Learning Techniques.

NetBeansJSPServletJavaScriptMYSQLApache TomcatOPENCV

SECRET MESSAGE SERVICE APP(Cryptex)

Mini Project-Vardhaman College | Aug 2016

> Created an Android App enabling encrypted messaging, decryptable by the intended recipient using a provided key.

Android StudioSQLite DatabaseAES Cryptographic AlgorithmsGoogle Play Store https://bit.ly/SKapp_Cryptex

PLAY MY TRACK (Android & Windows)

Independent Project-Google Play Store, Microsoft Store | Sept 2015

> Developed Music Player Apps with a Unique feature of playing two media files, one on each ear-bud of the pair of In-Ear earphone. for Android, Windows phones, Desktop. Using Android SDK, Java, Kotlin, C# .Net, Windows SDK.

Windows https://bit.ly/SKapp_PMTAndroid https://bit.ly/SKapp_PMTAMaterial DesignAdobe PhotoshopAndroid StudioVisual Studio

BLUEBUDDY - Wireless Chatting Without Internet

Project Exposition-CETA | Jan–Feb 2015

> Developed an Android app for Bluetooth messaging and can hence be used without Internet access.

Android StudioBluetooth APIsBluetoothSocketOpen Source ProjectGoogle Play Store http://bit.ly/SKapp_BlueBuddy

Education

Wayne State University

Master’s, Computer Science, AI Concentration | Detroit, MI

Aug. ’22 - May ’24

Relevant Coursework : Design and Analysis of Algorithms | Intelligent Systems : Algorithms and Tools | Data Mining : Algorithms and Applications | Principles of Web Technology | Principles of Cyber Security | Digital Image Processing & Analysis | Computer Graphics I | Machine Learning and Applications | Deep Learning for 3D Shapes and Scenes

Vardhaman College of Engineering

B.Tech, Computer Science & Engineering | Hyderabad, India

Sep. ’13 - Jun. ’17

Achievements

Apr 2017	Awarded as the “ <i>Best Outgoing Student</i> ” by department of CSE at Vardhaman College.
June’16	Published projects as products for Real-Time use on Microsoft Store & Google Play Store with 10K+ downloads.
Sept 2015	Won the first Prize for Windows Phone & Desktop app development in the Project Expo at Vardhaman.
Feb 2015	Achieved the first prize for Android Application Development at Vardhaman College.

Responsibilities

Nov’22- Sept’23	Graduate Research Assistant with Prof. Dr. Nic DePaula in tasks related to <u>LHEI</u> and his research around Government Administration by applying Data Analytics using Twitter, Facebook Web Scraped Data and Full Stack Development at WSU.
Jan-Apr’23	Lab Instructional Teaching Assistant for Data Structures and Algorithms using C++, CSC 2200 Computer Science II at WSU, provided teaching, contributed to course assessment, mentoring, and grading support to 60 students.
Aug’15-17	Was the college coordinator and a member of the departments organizing committee named CETA (<i>Computer Engineer’s Technical Association</i>).
Sept 2015	Organized Web Designing Workshop cum event at <i>TECHNOLITES’15</i> -Vardhaman’s annual techfest.

Technical Activities

Certifications

- > **Salesforce** *Platform Developer 1* .
- > **Microsoft Technology Associate:***Software Development Fundamentals*.
- > Certified **SAFe 5** Agile Practitioner.
- > Certified DBS **Software Security for Developers**.
- > Certified DBS **SRE** Practitioner.
- > **CISSO** : Certified Information Systems Security Officer.

Hackathons

- My team was shortlisted for the **ICICI Appathon** for creating new solutions in Banking and Financial Services among 3550 entries from all over India on HackerEarth.
- Participated in *Air Travel Hackathon by flydubai* and was among top 10 final teams at **T-hub, IIIT-Hyd**.
- Was among top 5 teams participated in “*Digital India-hackathon*” at **BITS Pilani Hyderabad** conducted by **Microsoft**.

Other

- Google AI Hands-On Event Sessions for The Rise of AI Agents & BuildwithAI
- Completed **ServiceNow** Administrator Training
- Worked on Independent learning POC using MERN Stack and other tech stack to upskill
- Worked on **SalesForce** POC projects using Apex Programming, Visualforce, Triggers, SOQL, Controllers, Workflows, Custom config, and APIs for the development of CRM solutions.
- Participated in **EPAM sponsored Global CodeRetreat** to celebrate passion and software crafting.
- Gave a paper presentation on “*Building Android Apps gets easier with Android Studio*” and a Seminar on “*Android Things*”.
- Attended a seminar on Cloud Computing (using IBM Bluemix) by **IBM**.
- Attended the **Windows App Development** workshop, by DevmenIT –a Microsoft partner company.
- Participated in *Battle Code –TechnoVoltz*, National level zonal competition by **Techfest- IIT Bombay**’s annual tech festival.
- Participated in *CodeKraft* and other events at the National level technical fest –Technolites, Vardhaman.